

22k-4785 - Syed Shaiq Hussain

22k-4782 – Mavia Mehmood

22k-4795 – Moiz Jadoon

DB Project Report

Project Title: Online Car Showroom Management System

Project Description

The Online Car Showroom Management System is a dynamic web application built using PHP and MySQL as the backend technologies and HTML, CSS for the frontend. It runs on the XAMPP server and aims to facilitate both customers and administrators in managing and viewing car-related information seamlessly. The system serves as a virtual showroom, allowing users to browse cars by categories and brands, send inquiries, and access detailed information about each car. Meanwhile, it provides administrative tools to manage cars, brands, categories, users, and inquiries.

Features

Customer Features

1. Home Page:

- Displays a list of all available cars.

- Customers can click on any car to view its details, including price, model, year, mileage, fuel type, and more.

2. Car Inquiry:

- Customers can send inquiries by providing their name, email, phone number, and a message.
- Inquiry data is stored in the inquiry_list table for admin and staff review.

3. Navigation Bar Options:

- **Categories:** Browse cars by category (e.g., SUV, Sedan, Hatchback).
- **Brands:** View cars listed under a specific brand.
- **About Us:** Provides details about the project, team members, and objectives.

Admin Features

1. Login Panel:

- Admin and staff can log in using secure credentials.
- Admins have full access, while staff has restricted access.

2. Car Management:

- Add new cars with details such as price, title, description, model, year, and features.
- Delete cars from the listing.
- Cars are saved in table named car_list.

3. Brand and Category Management:

- Add, update, or delete car brands and categories.
- These are stored in tables named brand_list and category_list.

4. User Management:

- Add or remove staff accounts.
- The users are stored in table named users.

5. Inquiry Management:

- View all customer inquiries with their details.
- Take necessary actions or respond to inquiries.

Staff Features

1. Restricted Inquiry Access:

- Staff can view only the inquiries sent by customers.

2. Login Panel:

- Staff members have secure login credentials.

Technologies Used

1. Frontend: HTML, CSS

2. Backend: PHP

3. Database: MySQL

4. Server: XAMPP

Normalization Report

The database was carefully normalized to ensure data integrity, eliminate redundancy, and improve performance. Below is a normalization breakdown:

1st Normal Form (1NF):

- **Definition:** Tables must contain atomic values, and each row must be unique.
- **Compliance:**
 - All tables (e.g., brand_list, car_list, category_list, inquiry_list, users) comply with 1NF.
 - Each column contains atomic values, and there are no repeating groups.

2nd Normal Form (2NF):

- **Definition:** The database must be in 1NF, and all non-key attributes must depend on the entire primary key.
- **Compliance:**
 - All tables are in 2NF:
 - car_list: Non-key attributes like price, mileage, and features are fully dependent on the id (primary key).
 - inquiry_list: Attributes like fullname and email are fully dependent on id.

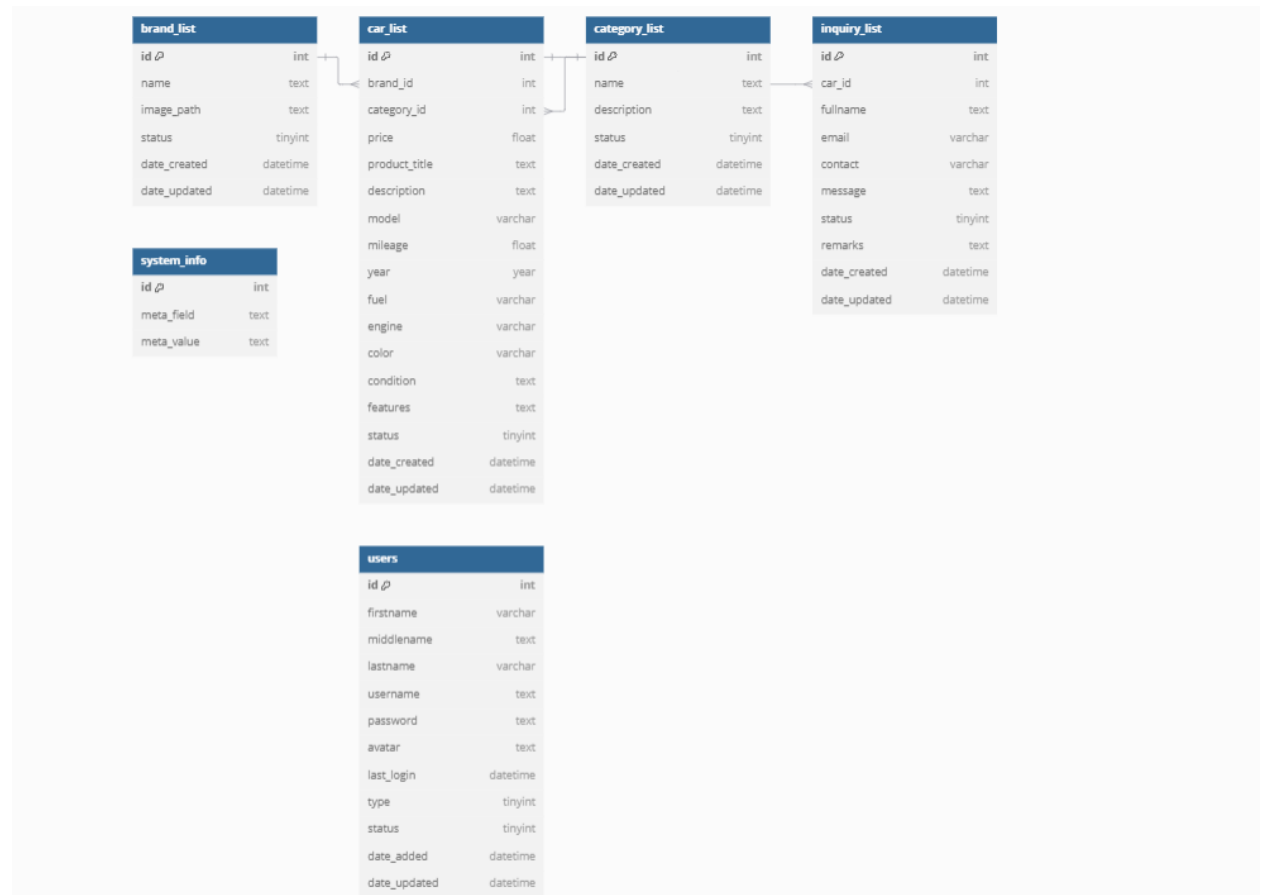
3rd Normal Form (3NF):

- **Definition:** The database must be in 2NF, and there should be no transitive dependencies.
- **Compliance:**
 - All tables meet 3NF:
 - car_list: No transitive dependencies between attributes.
 - inquiry_list: fullname, email, and other attributes depend directly on id.
 - Foreign keys like brand_id and category_id properly reference their respective tables.

Boyce-Codd Normal Form (BCNF):

- **Definition:** Every non-trivial functional dependency must involve a superkey.
- **Compliance:**
 - All tables satisfy BCNF as there are no dependencies where non-superkeys determine other attributes.

ER Diagram:



Conclusion

The Online Car Showroom Management System provides an efficient platform for customers and administrators to manage and explore car listings. It follows a well-structured database design, adhering to normalization principles up to BCNF. This ensures the application is robust, scalable, and free from redundancy, enhancing its usability and maintainability.